higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 Ear plugs
1.2 Carbon dioxide
1.3 Cullet
1.4 Ozone
1.5 Sasol

(5 × 1) **[5]**

QUESTION 2

2.1
2.1.1 True
2.1.2 False
2.1.3 False
2.1.4 True
2.1.5 False

(5 × 1) (5)

2.2
2.2.1 South African Nuclear Energy Corporation
2.2.2 Pretoria Portland Cement
2.2.3 South African Pulp and Paper Industries
2.2.4 Electrical Supply Commission
2.2.5 Transvaal Rubber Company

(5 × 1) (5)
[10]

QUESTION 3

3.1
- Copper
- Zinc
- Lead
- Iron
- Nickel
- Cobalt

(Any 4 × 1) (4)

3.2
- Mondi
- Sappi
- Nampak
- Kimberly-Clark

(Any 3 × 1) (3)

3.3
- Ethane
- Propane
- Butane

(3)

3.4 Underground mining is used when coal is found deep within the Earth's crust.✓ Deep shafts are dug underground with a network of tunnels leading to the coal beds.✓ The coal is removed by shoveling or explosives and loaded into carts or trucks and conveyed to the surface.✓ (3)
[13]

QUESTION 4

4.1 4.1.1
- Colourless
- Odourless
- Tasteless
- High surface tension
- High density
- Boils at 100 ˚C
- Freezes at 0˚C (Any 4 × 1) (4)

4.1.2 COD level is determined by measuring the amount of oxygen that is needed for water to react fully with potassium permanganate ($KMNO_4$). The general COD must be must be less than 75 *mg/l*. (2)

4.2
- Pre-treatment
- Primary sedimentation
- Activated sludge
- Disinfection
- Solid treatment (5)

4.3 Hardness of water refers to the dissolved salts in the water, √ which make it harder for soap to do effective cleaning.✓ Magnesium and calcium carbonates forms temporary hard water,✓ whereas magnesium and calcium of sulphates and chlorides are responsible for permanent hardness in water.✓ (4)

4.4 Chemical dosing is a process through which chemicals called flocculants✓ are added to water✓ in order to force suspended particles to stick to each other to form heavier lumps that settle easily.✓ Flocculants can either be in the form of a dry powder✓ or in solution form.✓ (5)

4.5
- Aluminium sulphate
- Ferric chloride
- Calcium oxide
- Calcium hydroxide
- Iron chloride (5)

4.6 By bubbling chlorine gas from the nozzle✓ in the bottom of the tank up through the water.✓ (2)

[27]

QUESTION 5

5.1
- Disinfectant
- Production of polyvinyl chloride plastics
- Manufacture of pharmaceutical products
- Cleaning (Any 3 × 1) (3)

5.2
- Job creation
- Cheaper food of a higher quality
- Cleaner water
- Production of products is cheaper
- Cheaper electricity
- Boosts economy of the country through exports
- Training of professionals (Any 5 × 1) (5)

5.3 5.3.1 Distillation is a process through which a mixture is separated into its pure components based on the difference in the boiling points of the components of the mixture.

5.3.2 It is caused by the accumulation of gases like CO_2 which traps heat in the atmosphere and causes the Earth's surface to become hotter. The extra heat is melting the polar ice caps, influencing the temperature and causing sea levels to rise.

(2 × 2) (4)

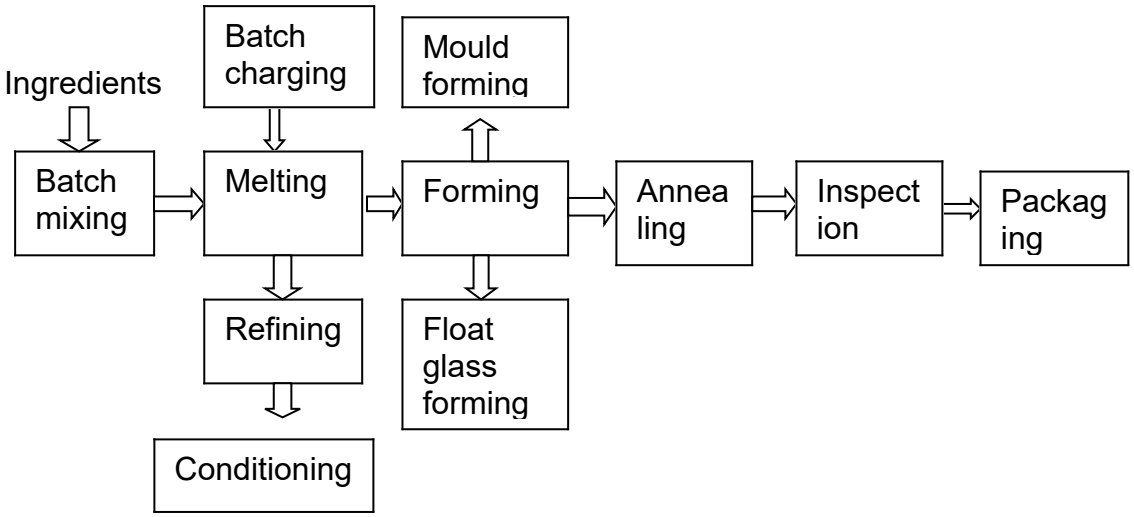
5.4 Refrigeration principle is based on the second law of thermodynamics which states that heat is transferred from a hotter to a cooler substance.✓ In a refrigerator, heat is removed from one side which is desired to be cooled and dumped on the other side.✓ The heat is removed by evaporating a liquid refrigerant, and the gas is transported outside the area that is desired to be cooled✓ and compressed until it condenses into a liquid again.✓ The boiling temperature of the refrigerant must be lower than the temperature that the fridge is desired✓ to be to avoid excessive electrical consumption.✓ (6)

[18]

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QUESTION 6

6.1



(ONE mark for each correct labelling.) (12)

6.2
- Silica
- Alumina
- Potassium oxide (3)

6.3
- Conduction
- Convection
- Radiation (3)

6.4
- Toxicity
- Viscosity
- Volatility (3)

6.5 6.5.1
- The disposal of chemical down the drain.
- The intentional evaporation of chemicals in a fume hood.
- The disposal of chemicals in a regular dustbin.

6.5.2
- Return it to the supplier.
- Re-use it on the site.
- Re-condition it; send it to the drum-registered re-conditioner or re-processors.
- Recycle it.
- Incinerate it.
- Create a hazardous waste landfill. (Any 3 × 1)

(2 × 3) (6)

[27]

TOTAL: 100